Spring 2021, MATH 8120 Real Analysis II, Final Exam

PRINT NAME:________________________________ **ID#:** ________________________

- Show your work. Answers with no or incorrect steps will not receive partial credit.
- The exam is self-explanatory. The instructor will not interpret the questions.

1. (5 points) Let Ω be an open bounded subset of $\mathbb{R}^n$ and $p \in (1, \infty)$. Let $\{f_k\}$ be a sequence of bounded functions in $L^p(\Omega)$ and $f_k \to f$ pointwisely for some $f \in L^p(\Omega)$. Show that $f_k \rightharpoonup f$ in $L^p(\Omega)$.

2. (5 points) Let X be a Banach space and let $\{f_k\}$ be a sequence in X^* that is weakly* convergent in X^*.

   (a) Show that $\{f_k\}$ is bounded.

   (b) Show that the weak* limit f of $\{f_k\}$ satisfies $\|f\| \leq \liminf_{k\to\infty} \|f_k\|$.

3. (5 points) Let $(X, \|\cdot\|)$ be a reflexive Banach space and let Z be a nonempty closed convex subset of X.

   (a) Show that, given any $x \in X$, there exists $y \in Z$ such that $\|x - y\| = \inf_{z \in Z} \|x - z\|$.

   (b) Show that y is unique if $(X, \|\cdot\|)$ is strictly convex.

4. (5 points) Let $(X, \langle \cdot, \cdot \rangle)$ be a Hilbert space and let $T \in L(X)$. Show that, if there exists a constant $\alpha > 0$ such that $\langle Tx, x \rangle \geq \alpha \|x\|^2$ for any $x \in X$, then T is bijective.

5. (5 points) Let X be a normed linear space and let $g : X \to \mathbb{R}$ be a convex and continuous function. Show that there exists $f \in X^*$ and $c \in \mathbb{R}$ such that $g(x) > f(x) + c$ for all $x \in X$.

6. (5 points) Let X be a Banach space and $x \in X$. Let $E_x := \{f \in X^* : \|f\| \leq \|x\|,\ f(x) = \|x\|^2\}$. Show that E_x is nonempty, closed, and convex

7. (5 points bonus) Mark each of the following assertions True (T) or False (F). Let X be a Banach space and $T \in L(X)$.

(a) $\square$ If $\{x_k\}$ is bounded in X, then it has a weakly convergent subsequence.

(b) $\square$ If $\{f_k\}$ is bounded in X^*, then it has a weakly* convergent subsequence.

(c) $\square$ The resolvent set $\rho(T)$ of T can be bounded in $\mathbb{C}$.

(d) $\square$ If $T^{-1}(X)$, then there exists $\delta > 0$ such that $S^{-1} \in L(X)$ for all S satisfying $\|T - S\| \leq \delta$.

(e) $\square$ If X^* is reflexive, then so is X.

All questions above are shown in this page.

1. (5 points) Let Ω be an open bounded subset of $\mathbb{R}^n$ and $p \in (1, \infty)$. Let $\{f_k\}$ be a sequence of bounded functions in $L^p(\Omega)$ and $f_k \to f$ pointwisely for some $f \in L^p(\Omega)$. Show that $f_k \rightharpoonup f$ in $L^p(\Omega)$.

2. (5 points) Let X be a Banach space and let $\{f_k\}$ be a sequence in X^* that is weakly* convergent in X^*.
   (a) Show that $\{f_k\}$ is bounded.
   (b) Show that the weak* limit f of $\{f_k\}$ satisfies $\|f\| \leq \liminf_{k\to\infty} \|f_k\|$.

3. (5 points) Let $(X, \|\cdot\|)$ be a reflexive Banach space and let Z be a nonempty closed convex subset of X.
   (a) Show that, given any $x \in X$, there exists $y \in Z$ such that $\|x - y\| = \inf_{z\in Z} \|x - z\|$.
   (b) Show that y is unique if $(X, \|\cdot\|)$ is strictly convex.

4. (5 points) Let $(X, \langle\cdot,\cdot\rangle)$ be a Hilbert space and let $T \in L(X)$. Show that, if there exists a constant $\alpha > 0$ such that $\langle Tx, x\rangle \geq \alpha\|x\|^2$ for any $x \in X$, then T is bijective.

5. (5 points) Let X be a normed linear space and let $g : X \to \mathbb{R}$ be a convex and continuous function. Show that there exists $f \in X^*$ and $c \in \mathbb{R}$ such that $g(x) > f(x) + c$ for all $x \in X$.

6. (5 points) Let X be a Banach space and $x \in X$. Let $E_x := \{f \in X^* : \|f\| \leq \|x\|,\ f(x) = \|x\|^2\}$. Show that E_x is nonempty, closed, and convex

7. (5 points bonus) Mark each of the following assertions True (T) or False (F). Let X be a Banach space and $T \in L(X)$.
   (a) ☐ If $\{x_k\}$ is bounded in X, then it has a weakly convergent subsequence.
   (b) ☐ If $\{f_k\}$ is bounded in X^*, then it has a weakly* convergent subsequence.
   (c) ☐ The resolvent set $\rho(T)$ of T can be bounded in $\mathbb{C}$.
   (d) ☐ If $T^{-1} \in L(X)$, then there exists $\delta > 0$ such that $S^{-1} \in L(X)$ for all S satisfying $\|T - S\| \leq \delta$.
   (e) ☐ If X^* is reflexive, then so is X.